

Machine Learning : to extract information and identify patterns.

for which tasks?

• **PREDICTION** :

• **INFERENCE** : we wish to know f , what relation there is between x and y .

Why do we think this works?
we assume it exists

$$Y = f(x_1, \dots, x_p) + \epsilon$$

noise

we ask it to be $\rightarrow E(\epsilon) = 0$
and $\text{Var}(\epsilon) = \sigma^2$

anche detto
Statistical Learning

VARIABLES $\rightarrow$ X / input variables / covariate / predictors / features
 $\rightarrow$ Y / output / response / target / label

TASKS

PREDICTION :

you have X , but not Y . Being $\langle \epsilon \rangle = 0$.
Then we can say that

$$\hat{Y} = \hat{f}(x)$$

when predicting Y knowing $x_1, \dots, x_p$

a **Black box model**

prediction for Y $\nearrow$
estimate for f $\nearrow$

how accurate is $\hat{Y}$ to Y ? it depends on

Reducible error
and
Irreducible error

depends on the model $\hat{f}$: in principle, we could reduce this by improving the model

depends on ϵ , which we cannot measure

$$\begin{aligned} E(Y - \hat{Y})^2 &= E[f(x) + \epsilon - \hat{f}(x)]^2 = E[f(x) - \hat{f}(x)]^2 + E[\epsilon^2] + 2E[\epsilon(f(x) - \hat{f}(x))] \\ &= E[f(x) - \hat{f}(x)]^2 + \text{Var}(\epsilon) \end{aligned}$$

Reducible error

Irreducible error

$\downarrow E(\epsilon^2) = \text{Var}(\epsilon)$
 $\downarrow E(\epsilon) = 0$

INFERENCE :

rather than getting a good accuracy on $\hat{Y}$, we would like to understand the relation between X and Y .

answering questions such as:

1) which predictors are important?

2) which predictors have a positive/negative relation with Y ?

3) is the relation "simple" (e.g. linear) or not?

Estimation of $\hat{f}$

- Say we have a number of points $n \rightarrow$ we call these points **TRAINING DATA** $\nearrow$ predictors $x_1, \dots, x_n$
- we want to find $\hat{f}$ such that $Y \approx \hat{f}(X)$ for every $(x, y)_{i=1}^n$ $\searrow$ labels $y_1, \dots, y_n$
- we do with

PARAMETRIC METHODS $\rightarrow$ ① assume a shape for f . The simplest shape? f is linear.
This model, whichever it is, will have **(p+1)** parameters, $\beta_0, \dots, \beta_p$
 $\hookrightarrow$ then the problem is to find the (p+1) parameters. how?

the fact that we jump from estimating f to estimating $\beta_0, \dots, \beta_p$ results in a distance between $\hat{f}$ and f .
To reduce this distance, we choose flexible models, that means more parameters. The downside of this is that we can adapt the model too much to the set of data we have

$\hookrightarrow$ for a linear model, the simplest approach is the least squares

NON-PARAMETRIC METHODS $\rightarrow$ there is no assumption on the shape of f , but they try to estimate f directly with the data points.
con: they need a lot of data points.
Here there is also the problem of overfitting.

Why should I prefer an approach to another? It depends on the questions I'm trying to answer.

Inference $\rightarrow$ a simple linear model is preferable to a model full of parameters, or with complicated shapes, and highlights the importance of single predictors.

Prediction $\rightarrow$ a good algorithm, very accurate, is preferable to a simple method, even though the algorithm becomes a "black box".

Supervised and Unsupervised Learning

- Supervised** $\rightarrow$ we have a response y_i for each x_i . $\rightarrow$ here we can look for relation between X and Y , and also perform prediction.
 - Unsupervised** $\rightarrow$ we have the predictors, but no response y_i . $\rightarrow$ here we can look for the relation between different x_i, x_j . $\rightarrow$ for example cluster analysis.
- also semi-supervised learning.

Regression and Classification problems

- quantitative response $\rightarrow$ regression
- qualitative response $\rightarrow$ classification

$\hookrightarrow$ the focus is on the response. The nature of the predictors is less important.

How good is a model?

- ↳ for example, how far is the predicted response to the actual response?
when working in **Regression** we have the **MEAN SQUARED ERROR (MSE)**

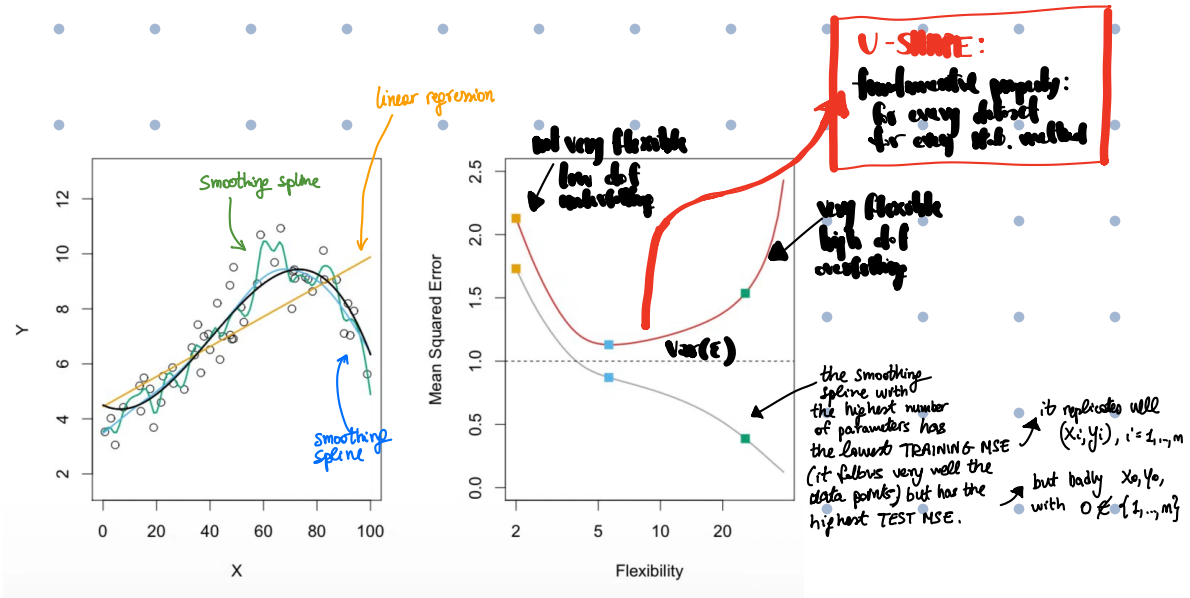
$$MSE = \frac{1}{n} \sum_{i=1}^m (y_i - \hat{f}(x_i))^2$$

• one wants MSE small

which data points?

those of a subset of the label data, called **TRAINING DATASET** $\Rightarrow$ **TRAINING MSE**.

- we need to use that training dataset to produce a model to fit well not the training dataset, but the future data points.



we need the model for which $\text{Ave} (y_i - \hat{f}(x_i))^2$ is smallest.